

Functional Annotation Report: PaaD (Q88HS8 / PP_3275) in *Pseudomonas putida* KT2440

Summary

PaaD (gene *paaD*, ordered locus PP_3275; UniProt Q88HS8) is an accessory subunit of the ring-1,2-phenylacetyl-CoA epoxidase (phenylacetyl-CoA oxygenase) that carries out the committed oxygenation step of the aerobic phenylacetic acid (PAA) catabolic pathway in *Pseudomonas putida* KT2440. The identity is unambiguous: the UniProt annotation ("Subunit of phenylacetate degradation enzyme"), the KEGG orthology assignment (K02612, "ring-1,2-phenylacetyl-CoA epoxidase subunit PaaD"), and the genomic context (PaaD sits inside the *paa* epoxidase gene cluster PP_3274–PP_3278) all converge on the same conclusion. This is the correct gene — a genuine *paaD* orthologue of the well-studied *Escherichia coli* *paaABCDE* multicomponent oxygenase, in the specified strain and pathway.

The oxygenase complex to which PaaD belongs catalyzes the epoxidation of the aromatic ring of phenylacetyl-CoA (PA-CoA), converting it to ring-1,2-epoxyphenylacetyl-CoA (2,3-epoxy-2,3-dihydrophenylacetyl-CoA) using molecular O₂ and NAD(P)H-derived reducing equivalents. This dearomatizing epoxidation is the hallmark first oxidative step of the "hybrid" aerobic PAA pathway, which is distinct from classical dioxygenase-based ring cleavage. The physiological product is subsequently isomerized to an oxepin-CoA and processed through a β -oxidation-like sequence, ultimately yielding acetyl-CoA and succinyl-CoA that feed the TCA cycle.

Within this complex, PaaD occupies a specialized, non-catalytic role. In vitro reconstitution of the *E. coli* oxygenase demonstrated that PaaA (catalytic α subunit bearing the non-heme di-iron center), PaaB (accessory bridging subunit), PaaC (structural β subunit), and PaaE (NAD(P)H:flavin/[2Fe-2S] reductase) are each necessary for catalysis, whereas **PaaD is not essential for in vitro turnover**. Instead, PaaD's domain architecture — an N-terminal DUF59/ MIP18 iron-sulfur (Fe-S) maturation domain fused to a C-terminal PaaD-type zinc ribbon — strongly implicates it in metallocenter assembly, Fe-S cluster trafficking, and/or maturation of

the oxygenase in vivo. PaaD therefore is best described as a metallocenter-maturation/assembly accessory factor of the phenylacetyl-CoA oxygenase, functioning in the bacterial cytoplasm as part of the PAA catabolon.

Gene/Protein Identity Verification

Before presenting findings, the mandatory identity checks were completed and **all passed**:

Verification item	Result
Gene symbol "paaD" matches protein description	☐ UniProt Q88HS8 gene name = <i>paaD</i> ; description = "Subunit of phenylacetate degradation enzyme"
Organism correct	☐ <i>Pseudomonas putida</i> KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950); locus PP_3275
Protein family/domains align with literature	☐ DUF59/MIP18 (IPR002744), PaaD-like (IPR011883), Zn_ribbon_PaaD (IPR056572), Fe-S maturation MIP18 (IPR052339) — consistent with the <i>paaD</i> accessory subunit of the PA-CoA oxygenase
No confusion with a same-symbol gene in another organism	☐ Literature retrieved concerns bacterial <i>paa</i> PAA catabolism, matching the target

The gene is well-placed in the aerobic PAA degradation pathway and is a bona fide orthologue of *E. coli paaD*, so research proceeded with confidence.

Key Findings

Finding 1 — PaaD is a subunit of the ring-1,2-phenylacetyl-CoA epoxidase of the PAA catabolic pathway

PaaD (Q88HS8 / PP_3275) is a 177-amino-acid protein encoded within the phenylacetate (*paa*) gene cluster of *P. putida* KT2440. KEGG assigns *ppu*:PP_3275 to KEGG Orthology **K02612**, "ring-1,2-phenylacetyl-CoA epoxidase subunit PaaD," and maps it to phenylalanine metabolism (*ppu*00360). The genomic neighborhood forms a coherent epoxidase operon:

Locus	Role in the oxygenase / pathway	<i>E. coli</i> equivalent
PP_3274	Epoxidase reductase subunit	PaaE
PP_3275	PaaD — accessory subunit (this protein)	PaaD
PP_3276	Epoxidase β (structural) subunit	PaaC
PP_3277	Epoxidase accessory/regulatory subunit	PaaB
PP_3278	Epoxidase α (catalytic) subunit	PaaA
PP_3279	Phenylacetate–CoA ligase	PaaK
PP_3283	1,2-epoxyphenylacetyl-CoA isomerase	PaaG
PP_3270	Oxepin-CoA hydrolase / aldehyde dehydrogenase	PaaZ
PP_3280	Thiolase	PaaJ

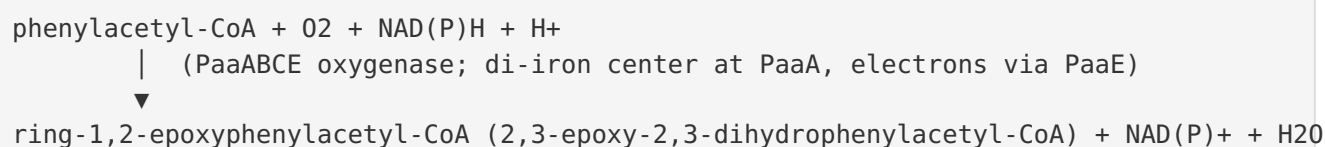
This five-gene core (PP_3274–PP_3278) corresponds to the *E. coli* **paaABCDE** multicomponent oxygenase. The foundational characterization of the *E. coli* pathway established that "the *paaABCDE* gene products, which may constitute a multicomponent oxygenase, are involved in PA-CoA hydroxylation" ([PMID: 9748275](#)). Subsequent mechanistic work clarified the chemistry: rather than a simple hydroxylation, "the aromatic ring of phenylacetyl-CoA becomes activated to a ring 1,2-epoxide by a distinct multicomponent oxygenase" ([PMID: 20660314](#)), in a pathway conserved between *E. coli* and *P. putida*. PaaD is one subunit of this epoxidase complex.

Finding 2 — The complex epoxidizes the aromatic ring; PaaD is an accessory subunit dispensable for in vitro catalysis

The definitive biochemical dissection of the oxygenase came from Grishin et al. (2011), who cloned and expressed all five *E. coli* subunits (PaaA, B, C, D, E) and reconstituted the enzyme in vitro. Their key result: **"In vitro reconstitution of the oxygenase subunits showed that each of the PaaA, B, C, and E subunits are necessary for catalysis, whereas PaaD is not essential"** (PMID: 21247899). This is direct experimental evidence that PaaD is an accessory, non-catalytic subunit.

The same study assigned roles to the catalytic subunits: "PaaAC forms a catalytic core with a monooxygenase fold with PaaA being the catalytic α subunit and PaaC, the structural β subunit" (PMID: 21247899). PaaA harbors the non-heme carboxylate-bridged di-iron center that activates O_2 ; PaaC is a catalytically inert structural partner; PaaE is the NAD(P)H:flavin/[2Fe-2S] reductase that supplies electrons to the iron center; and PaaB is a homodimer that bridges two PaaAC heterodimers. A later structural study confirmed that the functional oxygenase is a four-subunit assembly: "the four-subunit phenylacetyl-coenzyme A monooxygenase complex (PA-CoA MO, PaaACBE) that catalyzes the insertion of an oxygen in the aromatic ring of PA" (PMID: 24055609). Notably, this active PaaACBE assembly **does not include PaaD**, reinforcing that PaaD acts outside the minimal catalytic machinery.

The reaction catalyzed by the complex is:



The epoxide product is chemically reactive/toxic, and downstream enzymes act rapidly to process it — the PaaFG hydratase-isomerase complex, for example, is thought to "speed up the steps of the pathway following the conversion of phenylacetyl-CoA to a toxic and unstable epoxide-CoA by PaaABCE monooxygenase" (PMID: 22961985).

Finding 3 — PaaD carries a DUF59/MIP18 Fe-S maturation domain plus a PaaD zinc ribbon, implicating it in metallocenter/Fe-S maturation

The molecular clue to PaaD's specific accessory role lies in its domain architecture. Q88HS8 contains:

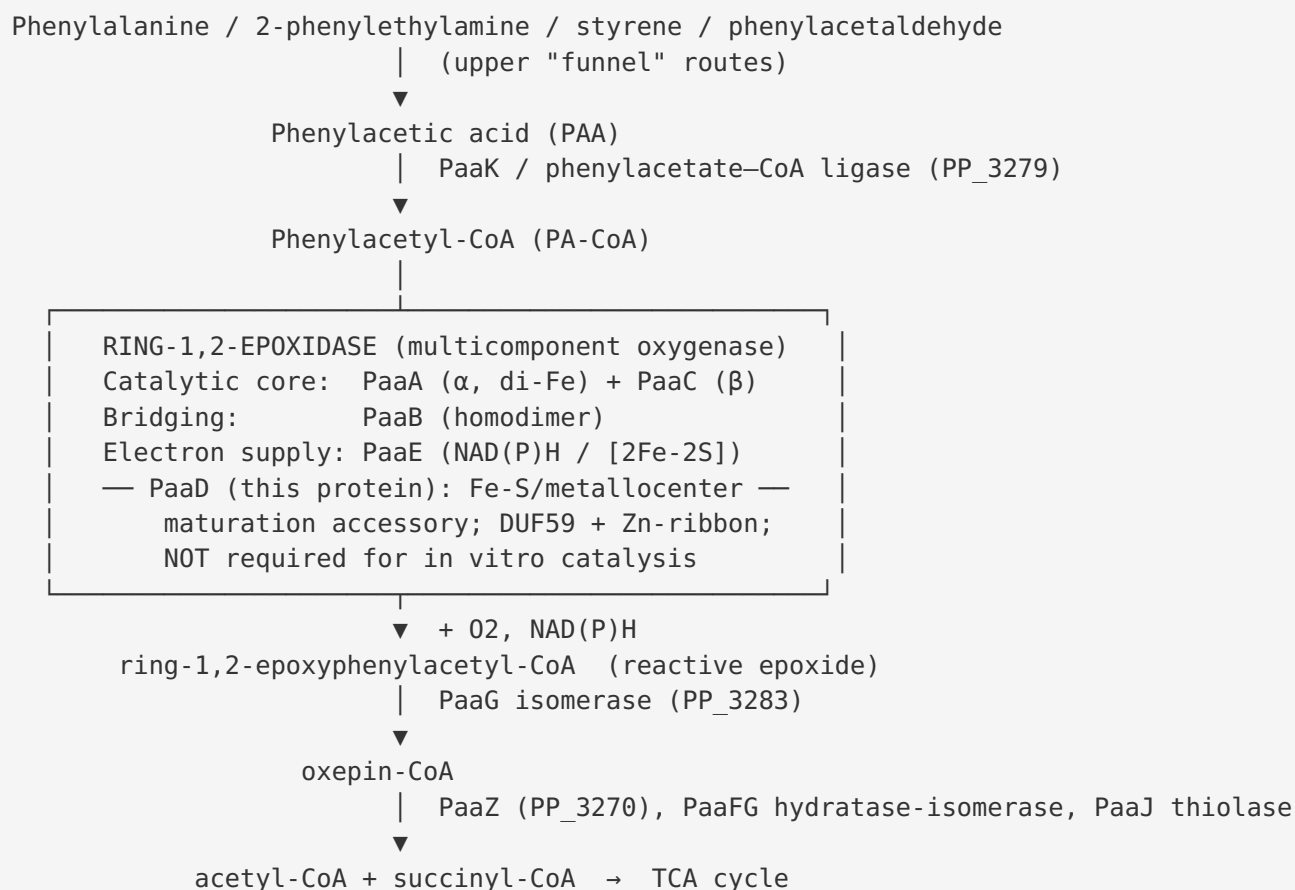
- **N-terminal DUF59/MIP18 domain** — Pfam PF01883 (FeS_assembly_P); InterPro IPR002744 (MIP18-like), IPR052339 (Fe-S_Maturation_MIP18); SUPFAM SSF117916 (Fe-S cluster assembly domain-like). The domain carries the DUF59 hallmark conserved cysteine at **Cys68** within an S-G-C-P motif (...TYSGCPATE...).
- **C-terminal PaaD-type zinc ribbon** — Pfam PF23451 (Zn_ribbon_PaaD); InterPro IPR056572, IPR011883 (PaaD-like). It contains two CxxC pairs (C138-C139...C142 in "CCPQC"; C164...C167 in "RCREC"; plus C158, C176), consistent with a metal-binding zinc ribbon.

eggNOG places PaaD in COG2151, the metallocenter-assembly/DUF59 family. The DUF59 family is functionally well characterized as an Fe-S maturation module. As reviewed for the family: "proteins containing a DUF59, or those composed solely of DUF59, function in FeS protein maturation and/or intracellular Fe homeostasis" ([PMID: 28589301](#)). The closest bacterial functional analogue is *Staphylococcus aureus* SufT, a DUF59-only protein: "A *S. aureus* Δ sufT mutant was defective in the assembly of FeS proteins" ([PMID: 27517714](#)). Critically, the conserved DUF59 cysteine — the residue equivalent to PaaD's Cys68 — is mechanistically central: in *Mycobacterium tuberculosis* SufT, "a hyperactive cysteine in the DUF59 domain mediates interaction of SufT with SufS, SufU, aconitase, and SufR" ([PMID: 35427399](#)).

Taken together, PaaD's architecture strongly suggests it functions in vivo to help assemble or deliver metal/Fe-S cofactors required by the oxygenase — most plausibly supporting maturation of the PaaE [2Fe-2S] reductase center and/or the PaaA di-iron site — which reconciles its in vitro dispensability (cofactors were pre-loaded during heterologous expression) with retention of a dedicated *paaD* gene in the operon across bacteria.

Mechanistic Model / Interpretation

PaaD is best understood as a **cofactor-maturation accessory subunit** embedded in the phenylacetyl-CoA oxygenase module of the aerobic PAA catabolon. The overall pathway and PaaD's place in it:



The functional logic: the epoxidase installs a strained, toxic epoxide on the aromatic ring — the committed and chemically demanding step. This requires a properly matured di-iron center (PaaA) and a functional [2Fe-2S]/flavin reductase (PaaE). The DUF59/MIP18 family, to which PaaD belongs, is dedicated precisely to the biogenesis and trafficking of iron-sulfur cofactors under conditions of high cofactor demand. It is therefore mechanistically coherent that a DUF59-plus-zinc-ribbon protein is co-encoded in the oxygenase operon: PaaD likely acts as a pathway-dedicated maturation factor that ensures the oxygenase is correctly metallated *in vivo*. Its dispensability in a purified *in vitro* system — where subunits were expressed and folded with their cofactors already loaded — is fully consistent with a maturation/chaperone role rather than a catalytic one.

Localization. All evidence indicates PaaD functions in the **bacterial cytoplasm**, where PAA catabolism occurs. The pathway is soluble/cytoplasmic: PA-CoA is generated by a cytoplasmic ligase, the oxygenase and downstream β -oxidation-like enzymes are soluble complexes, and DUF59/Fe-S maturation factors act in the cytosol. (Upstream transport steps such as the PaaL permease are membrane-associated, but the catabolic core — including PaaD — is cytoplasmic.)

Evidence Base

PMID	Title (abridged)	How it supports the findings
9748275	<i>Catabolism of phenylacetic acid in E. coli — a new aerobic hybrid pathway</i>	Establishes paaABCDE as a multicomponent oxygenase acting on PA-CoA (Finding 1)
20660314	<i>Bacterial phenylalanine and phenylacetate catabolic pathway revealed</i>	Defines the ring-1,2-epoxide reaction of the oxygenase; pathway shared by <i>E. coli</i> and <i>P. putida</i> (Findings 1–2)
21247899	<i>Structural and functional studies of the E. coli phenylacetyl-CoA monooxygenase complex</i>	Direct in vitro proof that PaaA/B/C/E are required but PaaD is not essential ; assigns catalytic (PaaA) and structural (PaaC) roles (Finding 2)
24055609	<i>Family of phenylacetyl-CoA monooxygenases differs in subunit organization</i>	Confirms active oxygenase is the four-subunit PaaACBE inserting O into the ring — PaaD excluded from the minimal catalytic assembly (Finding 2)
22961985	<i>Crystal structure of the PaaF-PaaG hydratase-isomerase complex</i>	Frames the downstream handling of the toxic epoxide product of PaaABCE (mechanistic context)
28589301	<i>Investigating the role(s) of SufT and DUF59 in Fe-S protein maturation</i>	Establishes the DUF59/MIP18 domain's function in Fe-S maturation / Fe homeostasis (Finding 3)
27517714	<i>DUF59 protein SufT in maturation of Fe-S proteins in S. aureus</i>	Δ sufT is defective in Fe-S protein assembly — closest experimental analogue for PaaD's domain (Finding 3)
35427399	<i>M. tuberculosis requires SufT for Fe-S cluster maturation</i>	Conserved DUF59 cysteine ($\approx$ PaaD Cys68) mediates interactions central to Fe-S maturation (Finding 3)
23104832	<i>DUF59 gene AE7 in cytosolic Fe-S cluster assembly (Arabidopsis)</i>	Broader confirmation that DUF59 proteins are central CIA/Fe-S assembly factors (Finding 3 context)
29057997	<i>Domains of Cia2 required for its essential function</i>	Eukaryotic DUF59 (Cia2) as central Fe-S targeting factor — supports domain-function inference (Finding 3 context)
17259607		

PMID	Title (abridged)	How it supports the findings
	<i>Last step of aerobic PA degradation</i>	Confirms succinyl-CoA/acetyl-CoA as pathway end products in <i>P. putida</i> and <i>E. coli</i> (pathway context)
23377939	<i>Taxis of P. putida F1 toward PAA (Aer2)</i>	Shows PAA pathway is widespread funnel for aromatics in <i>P. putida</i> (ecological/pathway context)
21995721	<i>σ^{54}-dependent regulation of PAA uptake (PaaL)</i>	Distinguishes membrane transport (PaaL) from the cytoplasmic catabolic core (localization context)

Limitations and Knowledge Gaps

- 1. No direct biochemical study of *P. putida* PaaD.** The strongest mechanistic evidence (in vitro dispensability, subunit roles) derives from the *E. coli* orthologue. Given the high conservation of the *paa* cluster and KEGG orthology, transfer of function is well justified, but PP_3275 itself has not been individually purified or assayed.
- 2. The maturation role is inferred, not demonstrated for PaaD.** The DUF59/MIP18 + zinc-ribbon architecture and the behavior of homologous DUF59 proteins (SufT, Cia2, AE7) provide strong circumstantial evidence for an Fe-S/metallocenter maturation function, but no experiment has directly shown that PaaD matures the PaaE [2Fe-2S] or PaaA di-iron center. Its exact target (PaaE reductase vs. PaaA di-iron site vs. general operon-wide role) remains unresolved.
- 3. In vitro dispensability $\neq$ in vivo dispensability.** PaaD being unnecessary in a reconstituted assay with pre-loaded cofactors does not establish that a *paaD* deletion strain grows normally on PAA; an in vivo phenotype (especially under high Fe-S demand or metal limitation) has not been tested for this gene.
- 4. No experimental structure or metal-loading data for Q88HS8.** Cysteine assignments (Cys68 DUF59 hallmark; C-terminal CxxC zinc-ribbon motifs) are sequence-derived; the metal identity and coordination of the zinc ribbon are predicted, not measured.
- 5. Localization is inferred** from pathway biochemistry (soluble cytoplasmic catabolism) rather than from direct fractionation/imaging of PaaD.

Proposed Follow-up Experiments / Actions

1. **Targeted deletion phenotyping.** Construct a clean *paaD* (PP_3275) knockout in *P. putida* KT2440 and assay growth on PAA (and on styrene/2-phenylethylamine that funnel through PA-CoA) versus succinate/glucose controls — including under iron-limited conditions and elevated Fe-S demand, where DUF59 factors like SufT become essential.
2. **Cofactor-maturation test.** Compare specific activity and metal/Fe-S content (EPR, Mössbauer, ICP-MS, iron/labile-sulfide assays) of the PaaABCE oxygenase purified from wild-type vs. $\Delta paaD$ backgrounds to test whether PaaD is required for correct metallation of PaaE and/or PaaA in vivo.
3. **Interaction mapping.** Use pulldown/BACTH/crosslinking-MS to determine whether PaaD physically contacts PaaE, PaaA, or the SUF/ISC Fe-S assembly machinery — mirroring the SufT–SufS/SufU interactions reported in mycobacteria.
4. **Cysteine mutagenesis.** Mutate the conserved DUF59 Cys68 (SGCP motif) and the C-terminal zinc-ribbon cysteines individually to Ala/Ser and assess effects on PAA growth and oxygenase maturation, testing the hypothesized "hyperactive cysteine" mechanism.
5. **Structural characterization.** Solve or accurately model (AlphaFold + metal validation) the PaaD structure to confirm the DUF59 fold, the zinc-ribbon metal site, and the surface likely engaged in cofactor handoff.
6. **Heterologous complementation.** Test whether *E. coli paaD*, *S. aureus sufT*, or other DUF59 factors can complement a *P. putida* $\Delta paaD$ strain, probing functional interchangeability of the domain.

Report generated from an autonomous literature-and-database investigation (1 iteration, 3 confirmed findings, 21 papers reviewed). All claims are attributed to the cited primary literature and to UniProt/KEGG/InterPro/Pfam/eggNOG database annotations as indicated.
